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Experiment Title : Demonstration of Router Hardware and Connecting Different LANs

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Experiment No. 07: Demonstration of Router Hardware and Connecting Different LANs

Aim

To identify the hardware interfaces of a Router and to design, configure, and verify the connectivity between two distinct Local Area Networks (LANs) using a Router in Cisco Packet Tracer.

Theory

What is a Network Router?
A network router is a networking device that connects two or more different networks together. It operates at the Network Layer (Layer 3) of the OSI model. A router forwards data packets between networks based on IP addresses. It is commonly used to connect a Local Area Network (LAN) to the Internet. Routers help determine the best path for data transmission.

Explain the role of a "Default Gateway" in a network.
A default gateway is the IP address of the router that connects a local network to other external networks. When a device wants to send data outside its own network, it forwards the packet to the default gateway. The router then routes the packet to the appropriate destination network. Without a default gateway, devices can only communicate within their local network. It acts as the exit point for outgoing traffic.

How does a Router differ from a Switch?
A router connects different networks, while a switch connects devices within the same network. A router uses IP addresses to forward packets, whereas a switch uses MAC addresses. Routers operate at Layer 3 (Network Layer), and switches operate at Layer 2 (Data Link Layer). Routers can perform functions such as network address translation (NAT) and firewall filtering, while switches mainly manage local data traffic. Therefore, routers enable inter-network communication, while switches manage intra-network communication.

What is a Routing Table and how is it used to forward packets?

A routing table is a data table stored inside a router that contains information about available network paths. It includes destination networks, next-hop addresses, and interface details. When a packet arrives, the router checks the destination IP address against the routing table. The router then selects the best possible route and forwards the packet through the appropriate interface. The routing table ensures efficient and accurate packet delivery.

Hardware Components Required

- **PC-PT (x6):** End devices acting as network nodes (3 for LAN 1, 3 for LAN 2).
- **Switch 2960 (x2):** Two 24-port switches to create independent local networks. One switch for each LAN.
- **Router ISR 4321 (x1):** The device to connect the two LANs.

- **Cabling:** Copper Straight-Through cables for all connections.

Hardware and Connection Setup

- LAN 1 (Network ID): 192.168.1.0 | Subnet Mask: 255.255.255.0
- LAN 2 (Network ID): 10.0.0.0 | Subnet Mask: 255.0.0.0
- PC Naming Convention:
  - LAN 1: S11_01_A, S11_01_B, S11_01_C (IPs: .2, .3, .4)
  - LAN 2: S11_01_D, S11_01_E, S11_01_F (IPs: .2, .3, .4)

Procedure

Step 1: Physical Topology Design

- Place the ISR 4321 Router in the center of the workspace.
- Place two 2960 Switches (one for each LAN) and six PCs (three per switch).
- Naming: Rename PCs following the convention (e.g., S11_01_A through S11_01_F).

Step 2: Hardware Identification

- Click on the Router and select the Physical tab.
- Observe the modular slots and the GigabitEthernet ports.
- Note the power switch; ensure the router is turned ON.

Step 3: Physical Cabling

- Connect Switch 0 to Router interface GigabitEthernet 0/0/0.
- Connect Switch 1 to Router interface GigabitEthernet 0/0/1.
- Connect all PCs to their respective switches.
- Observation: Note that the link lights between the switches and the router are Red. This is because router ports are disabled by default.

Step 4: Router Interface Configuration (The "Bridge" Setup)

1. Click **Router > CLI**. Type ‘no’ at the initial setup prompt.
2. Enter the following commands to configure the gateways:
   - Router> enable

- Router# config t
- Router(config)# interface gigabitEthernet 0/0/0
- Router(config-if)# ip address 192.168.1.1 255.255.255.0
- Router(config-if)# no shutdown
- Router(config-if)# interface gigabitEthernet 0/0/1
- Router(config-if)# ip address 10.0.0.1 255.0.0.0
- Router(config-if)# no shutdown

Observation: The link lights should now turn Green.

Command	Purpose / Function
enable	Moves the user from **User EXEC Mode** (view only) to **Privileged EXEC Mode**. This allows the user to view detailed configurations and enter setup modes.
config t (configure terminal)	Moves the router into **Global Configuration Mode**. This is the only mode where you can make changes that affect the router as a whole.
interface gigabitEthernet 0/0/0	Moves the router into **Interface Configuration Mode**. It tells the router, "The next commands I type are only for this specific physical port."
ip address [IP] [Mask]	Assigns a logical Identity (IP) and defines the network size (Subnet Mask) for that interface. This becomes the **Default Gateway** for that LAN.
no shutdown	By default, Cisco router ports are "Administratively Down" (turned off for security). This command **powers on** the interface to allow data flow.

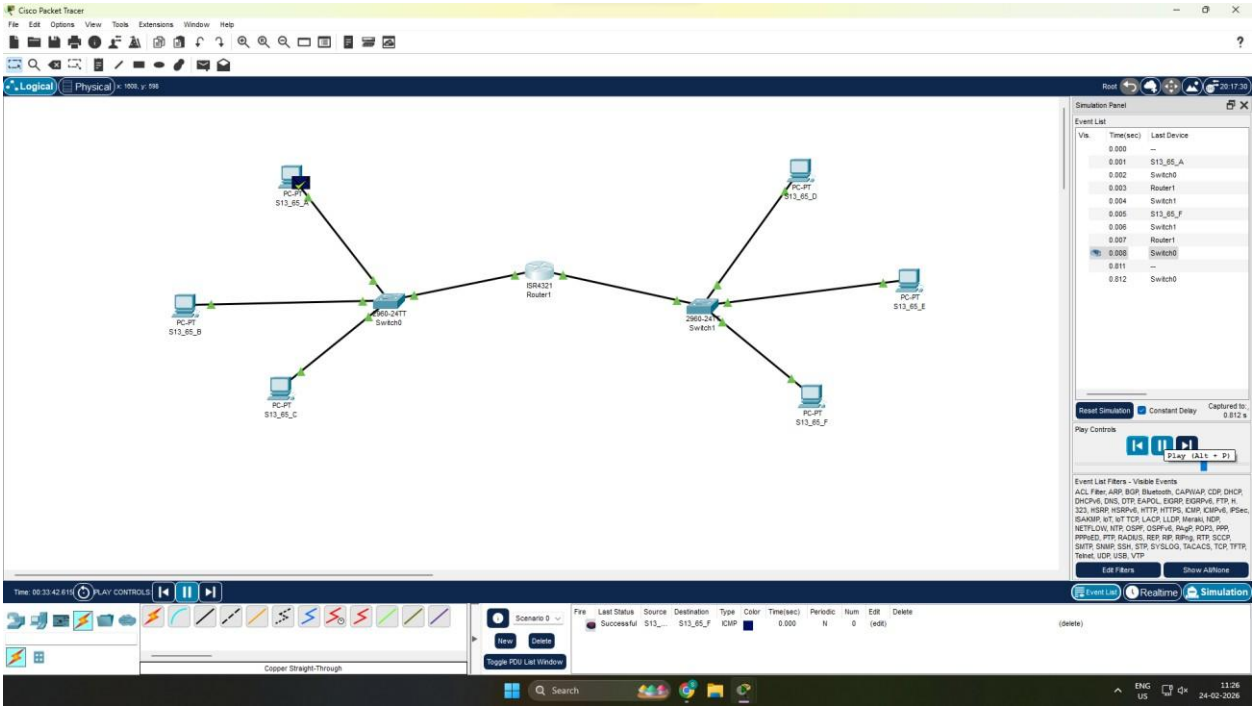
Step 5: IP Addressing (Logical Setup)

- **For LAN 1 PCs:** Set IPs 192.168.1.x and set Default Gateway to 192.168.1.1.
- **For LAN 2 PCs:** Set IPs 10.0.0.x and set Default Gateway to 10.0.0.1.

Step 6: Testing and Simulation

1. **Real-time:**
   - Open Command Prompt on **S11_01_A** and type ping 10.0.0.4.
   - Verify connectivity. (Note: The first ping may timeout due to ARP).
2. **Simulation Mode:**

○ Filter for ICMP. Use the Add Simple PDU (Envelope) to click S11_01_A (LAN 1) then S11_01_F (LAN 2). ○ Observe the packet crossing from the 192.x network to the 10.x network through the router.



Observations

- **Gateway Necessity:** If the "Default Gateway" is not configured on the PCs, the packet never reaches the router and communication fails.
- **Routing Decision:** The Router stops the broadcast from LAN 1 and intelligently forwards the packet specifically to the interface connected to LAN 2.
- **Different Subnets:** Communication was successfully established between two completely different IPs, which is impossible with only a switch.

Conclusion

The experiment successfully demonstrates that a **Router** is the essential hardware for Inter-LAN or Inter-Network communication. It proves that while switches manage local traffic, routers are required to cross network boundaries by using logical IP addresses.